

$$\begin{aligned}
& \left(-\frac{2}{27} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_d^{\mathbf{p}}] \delta_{i1i2} + \frac{5}{36} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_d^{\mathbf{p}}] \delta_{i1i2} - \frac{8}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_d^{\mathbf{p}}] \delta_{i1i2} - \right. \\
& \frac{2}{9} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_e^{\mathbf{p}}] \delta_{i1i2} + \frac{5}{12} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_e^{\mathbf{p}}] \delta_{i1i2} - \frac{8}{45} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_e^{\mathbf{p}}] \delta_{i1i2} - \\
& \frac{1}{9} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_l^{\mathbf{p}}] \delta_{i1i2} + \frac{5}{24} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_l^{\mathbf{p}}] \delta_{i1i2} - \frac{4}{45} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_l^{\mathbf{p}}] \delta_{i1i2} - \\
& \frac{1}{27} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_q^{\mathbf{p}}] \delta_{i1i2} + \frac{5}{72} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_q^{\mathbf{p}}] \delta_{i1i2} - \frac{4}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_q^{\mathbf{p}}] \delta_{i1i2} - \\
& \frac{8}{27} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_u^{\mathbf{p}}] \delta_{i1i2} + \frac{5}{9} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_u^{\mathbf{p}}] \delta_{i1i2} - \frac{32}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_u^{\mathbf{p}}] \delta_{i1i2} - \\
& \frac{1}{9} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} - \frac{1}{6} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} + \frac{8}{45} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{12} g_1^4 LF_{2,1,0}[m_1, m_e^{i1}] \delta_{i1i2} - \frac{1}{12} g_1^4 LF_{2,2,-1}[m_1, m_e^{i1}] \delta_{i1i2} + \\
& \frac{1}{6} g_1^4 LF_{3,1,-1}[m_1, m_e^{i1}] \delta_{i1i2} - \frac{1}{12} g_1^4 LF_{4,1,-2}[m_1, m_e^{i1}] \delta_{i1i2} - \frac{1}{12} g_1^4 LF_{2,1,0}[m_1, m_e^{i2}] \delta_{i1i2} - \\
& \frac{1}{12} g_1^4 LF_{2,2,-1}[m_1, m_e^{i2}] \delta_{i1i2} + \frac{1}{6} g_1^4 LF_{3,1,-1}[m_1, m_e^{i2}] \delta_{i1i2} - \frac{1}{12} g_1^4 LF_{4,1,-2}[m_1, m_e^{i2}] \delta_{i1i2} - \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} LF_{2,2,0}[m_d^r, m_q^{\mathbf{p}}] \delta_{i1i2} + \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} LF_{3,2,-1}[m_d^r, m_q^{\mathbf{p}}] \delta_{i1i2} - \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} LF_{2,2,0}[m_e^r, m_l^{\mathbf{p}}] \delta_{i1i2} + \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} LF_{3,2,-1}[m_e^r, m_l^{\mathbf{p}}] \delta_{i1i2} + \\
& \frac{1}{6} g_1^4 LF_{2,1,0}[m_e^{i1}, m_1] \delta_{i1i2} - \frac{1}{12} g_1^4 LF_{3,1,-1}[m_e^{i1}, m_1] \delta_{i1i2} + \frac{1}{6} g_1^4 LF_{2,1,0}[m_e^{i2}, m_1] \delta_{i1i2} - \\
& \frac{1}{12} g_1^4 LF_{3,1,-1}[m_e^{i2}, m_1] \delta_{i1i2} + \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} LF_{3,1,0}[m_l^{\mathbf{p}}, m_e^r] \delta_{i1i2} + \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} LF_{3,2,-1}[m_l^{\mathbf{p}}, m_e^r] \delta_{i1i2} - \frac{3}{4} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} LF_{4,1,-1}[m_l^{\mathbf{p}}, m_e^r] \delta_{i1i2} + \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} LF_{5,1,-2}[m_l^{\mathbf{p}}, m_e^r] \delta_{i1i2} + \frac{1}{6} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{2,1,0}[m_l^{\mathbf{p}}, \tilde{\mu}] - \\
& \frac{1}{12} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{2,2,-1}[m_l^{\mathbf{p}}, \tilde{\mu}] - \frac{1}{12} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{3,1,-1}[m_l^{\mathbf{p}}, \tilde{\mu}] + \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} LF_{3,1,0}[m_q^{\mathbf{p}}, m_d^r] \delta_{i1i2} + \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} LF_{3,2,-1}[m_q^{\mathbf{p}}, m_d^r] \delta_{i1i2} - \\
& \frac{5}{4} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} LF_{4,1,-1}[m_q^{\mathbf{p}}, m_d^r] \delta_{i1i2} + g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} LF_{5,1,-2}[m_q^{\mathbf{p}}, m_d^r] \delta_{i1i2} + \\
& \frac{1}{6} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} LF_{2,2,0}[m_q^{\mathbf{p}}, m_u^r] \delta_{i1i2} - \frac{1}{12} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} LF_{3,2,-1}[m_q^{\mathbf{p}}, m_u^r] \delta_{i1i2} - \\
& \frac{1}{6} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} LF_{3,1,0}[m_u^r, m_q^{\mathbf{p}}] \delta_{i1i2} - \frac{1}{6} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} LF_{3,2,-1}[m_u^r, m_q^{\mathbf{p}}] \delta_{i1i2} - \\
& \frac{1}{2} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} LF_{4,1,-1}[m_u^r, m_q^{\mathbf{p}}] \delta_{i1i2} + g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} LF_{5,1,-2}[m_u^r, m_q^{\mathbf{p}}] \delta_{i1i2} + \\
& \frac{5}{12} g_1^4 LF_{4,1,-2}[\tilde{\mu}, m_1] \delta_{i1i2} - \frac{1}{3} g_1^4 LF_{5,1,-3}[\tilde{\mu}, m_1] \delta_{i1i2} + \frac{5}{4} g_1^2 g_2^2 LF_{4,1,-2}[\tilde{\mu}, m_2] \delta_{i1i2} - \\
& g_1^2 g_2^2 LF_{5,1,-3}[\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{12} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{2,1,0}[\tilde{\mu}, m_l^{\mathbf{p}}] + \\
& \frac{5}{12} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{3,1,-1}[\tilde{\mu}, m_l^{\mathbf{p}}] - \frac{1}{6} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{4,1,-2}[\tilde{\mu}, m_l^{\mathbf{p}}] + \\
& \frac{1}{8} g_1^4 LF_{2,1,1,-1}[m_1, m_e^{i1}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{4} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_e^{i1}, \tilde{\mu}] \delta_{i1i2} + \\
& \frac{1}{8} g_1^4 LF_{2,1,1,-1}[m_1, m_e^{i2}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{4} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_e^{i2}, \tilde{\mu}] \delta_{i1i2} + \\
& \frac{1}{8} g_1^4 LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_e^{i1}] \delta_{i1i2} - \frac{1}{8} g_1^4 m_1^2 LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^{i1}] \delta_{i1i2} + \\
& \frac{1}{8} g_1^4 LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_e^{i2}] \delta_{i1i2} - \frac{1}{8} g_1^4 m_1^2 LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^{i2}] \delta_{i1i2} - \\
& \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^{\mathbf{p}}] - \frac{3}{8} g_2^2 \tilde{\mu}^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^{\mathbf{p}}] + \\
& \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_l^{\mathbf{p}}] + \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{2,1,1,0}[\tilde{\mu}, m_2, m_l^{\mathbf{p}}] + \\
& \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} LF_{1,1,1,1,0}[m_1, m_e^{i1}, m_l^{\mathbf{p}}, \tilde{\mu}] - \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \\
& LF_{2,1,1,1,-1}[m_1, m_e^{i1}, m_l^{\mathbf{p}}, \tilde{\mu}] + \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y}_$$